PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Caesarean Sections in Teaching Hospitals: Systematic Review and Meta-analysis of Hospitals in 22 Countries
AUTHORS	Hoxha, Ilir; Zhubi, Esra; Grezda, Krenare; Kryeziu, Blerta; Bunjaku, Jeta; Sadiku, Fitim; Agahi, Riaz; Lungu, Daniel; Bonciani, Manila; Little, George

VERSION 1 – REVIEW

REVIEWER	Jose Guilherme Cecatti Cecatti Department of Obstetrics and Gynecology School of Medical Sciences University of Campinas Campinas, Brazil
REVIEW RETURNED	13-Jul-2020

GENERAL COMMENTS	This is a well planned and performed systematic review on the prevalence of Cesarean sections in teaching hospitals in comparison with no teaching hospitals. While I do recognize the quality of the manuscript and the procedures adopted for this systematic review that has merit for being published, there are some important aspects that need more detailed information before it can be formally accepted. Initially, there are two important topics I want to highlight. The first refers to the appropriateness of approaching this topic with a systematic review. I personally think that a compilation of several routine data on births or databases from several studies that could be directly assessed under permission would result in strong evidence, considering the characteristic of each hospital could be better evaluated. And perhaps a more powerful ipd-metanalysis could also be performed. However, of course, I am not proposing to change completely the scope of the manuscript and to perform a new study. But I would appreciate a comment on this included in the discussion session. The second point, the most important, refers to what is defined or operationally understood as "teaching hospital". I guess the simple information from an article stating that those data come from a teaching hospital is not enough to classify it as such. In fact, the definition used in the study was presented only on page 6, lines 45-52, and says that "The teaching hospitals were considered hospitals with academic centre/teaching status, e.e. hospitals where residency training and/or academic research takes place...". This definition should come first, in the abstract, and also at the beginning of the methods session because it is really important. I suspect that there is a high likelihood of selection bias of studies based only on the information from the article of this status for each hospital. This point I judged as so important that it should be considered along with all the manuscript, and mainly referred to in the discussion session as a

	possible limitation of the study. As an example, I know hospitals that could be simultaneously classified as both, teaching and no teaching hospital, depending on the patients´ insurance and medical staff in charge of them. Are the authors sure this is not the condition for any of the hospitals included? Following there are other points needing some changes or adaptations: Abstract, page 2: apart from what has already been said regarding the definition of a teaching hospital, it lacks information on the number of hospitals and women the SR used to generate the information presented. In the conclusion session, the phrase "Nevertheless, teaching hospitals, in several subgroups, have a lower or higher likelihood of cesarean sections" are more or less obvious after the main conclusion and seems useless. Introduction, pages 4-5: well structured, however, I would recommend introducing the topic of the definition of teaching hospital also here. In addition, another argument regarding the possible characteristics of teaching hospitals should be better explored, on the more appropriate adherence to evidence-based guidelines or clinical protocols in these settings. Material & Methods, page 5-7: As already said, the definition of teaching hospital should come first. There is no information at all on language restriction. How was this treated? I realized that some other databases were not checked like for instance ScieLO. On page 5, line 52, it is said "Studies were excluded to remove overlapping data" and this is not very clear. Does it refer to the situation when more than one published article reported the same dataset of cases? Please clarify. Discussion and Conclusion: they are in accordance with what was planned and presented. Please consider the comments above to adapt them accordingly.

REVIEWER	Mohammed Walid Zimmo Department of Obstetrics and Gynecology, Al Shifa medical complex, Gaza, Palestine
REVIEW RETURNED	11-Sep-2020

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. It is very interesting to compared the odds of caesarean section in teaching hospitals compared with non-teaching hospitals. I commend the authors for this important contribution. This is an interesting and important topic, and a well-performed study covering a large population in many countries over an extended period of time could add to the understanding of changes in obstetric practice, and possible strategies to avoid unnecessary cesarean sections. The authors had chosen a significant methodology and well presenting the results. I have small comment in the introduction lines 10, authors state “The WHO in 1985 suggested an optimal rate of 10-15%" This recommendation was done by WHO in 1983 and they change it in 2015. WHO recommends delivery by cesarean section when clinically indicated. Please revise this reference: https://apps.who.int/iris/bitstream/handle/10665/161442/WHO_RHR_15.02_eng.pdf;jsessionid=DB7E8077D62D77B43FEB6BA19049A60F?sequence=1 I agree with your discussion specially with your statement "In order to improve the quality of healthcare provision for deliveries, the interplay of factors associated with teaching status of hospitals must be examined, as well as outcomes, related to caesarean section rates but also other outcomes such as proportion of morbidity, mortality, poor-perinatal outcomes and particular forms of delivery (i.e. emergency and primary caesarean, vaginal birth after caesarean)" since we miss sufficient studies which explore proportion of morbidity, mortality, poor-perinatal outcomes.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Jose Guilherme Cecatti Cecatti
Institution and Country: Department of Obstetrics and Gynecology
School of Medical Sciences
University of Campinas
Campinas, Brazil
Competing interests: None declared
Comments to the Author
Comment: This is a well planned and performed systematic review on the prevalence of Cesarean sections in teaching hospitals in comparison with no teaching hospitals. While I do recognize the quality of the manuscript and the procedures adopted for this systematic review that has merit for being published, there are some important aspects that need more detailed information before it can be formally accepted.
Responset: Thank you for your kind words and for the constructive feedback. This helped us to address some of omissions in first draft and improves the clarity and quality of our study.

Comment: Initially, there are two important topics I want to highlight. The first refers to the appropriateness of approaching this topic with a systematic review. I personally think that a compilation of several routine data on births or databases from several studies that could be directly assessed under permission would result in strong evidence, considering the characteristic of each hospital could be better evaluated. And perhaps a more powerful ipd-metanalysis could also be performed. However, of course, I am not proposing to change completely the scope of the manuscript and to perform a new study. But I would appreciate a comment on this included in the discussion session.

Response: Thank you for your suggestion. We fully agree that using individual patient data can provide way a more precise estimation of effects and adjustment. Be that in context specific analysis (at country, regional even hospital level) or through individual patient data (IPD) meta-analysis. Nevertheless, I think our meta-analysis will all detailed analysis of papers and other synthesised information provide a valuable resource for such efforts and we state that in implications. Among other, it also provides material for a conceptual framework, ideas for outcome definition, effect variables as well as adjusting factor to consider. To address your comment, we integrated IPD meta-analysis as additional option for future research in implications section of discussion. We specifically state: “It may also provide an impetus for further research, particularly setting-specific studies, or even individual patient data meta-analysis. Such studies would allow a more precise and in-depth analysis of the characteristics of hospitals associated with teaching status, such as resource availability or adherence to clinical guidelines.” (Line 268-272)

Comment: The second point, the most important, refers to what is defined or operationally understood as "teaching hospital". I guess the simple information from an article stating that those data come from a teaching hospital is not enough to classify it as such. In fact, the definition used in the study was presented only on page 6, lines 45-52, and says that "The teaching hospitals were considered hospitals with academic centre/teaching status, e.e. hospitals where residency training and/or academic research takes place...". This definition should come first, in the abstract, and also at the beginning of the methods session because it is really important. I suspect that there is a high likelihood of selection bias of studies based only on the information from the article of this status for each hospital. This point I judged as so important that it should be considered along with all the manuscript, and mainly referred to in the discussion session as a possible limitation of the study. As an example, I know hospitals that could be simultaneously classified as both, teaching and no teaching hospital, depending on the patients´ insurance and medical staff in charge of them. Are the authors sure this is not the condition for any of the hospitals included?
Response: Many thanks for this suggestion. We agree that we should have been clearer with definition. We have now added more details about teaching hospital definition using references in methods section of manuscript. (Line 136-148) For more, based on these definitions, we have created new classification and grouping of studies (Line 149-153) and performed again the meta-analysis. As a result of this we have made substantial modifications in manuscript. (Figure 2, Figure 3, Online Appendix, and manuscript) Another option would have been to only perform subgroup analysis. That would have been easier for us. We think re-doing meta-analysis with new grouping of studies provides a more precise estimation of teaching effect of hospitals and addresses some of limitations of our initial analysis which you have noted.

Comment: Following there are other points needing some changes or adaptations:
Abstract, page 2: apart from what has already been said regarding the definition of a teaching hospital, it lacks information on the number of hospitals and women the SR used to generate the information presented.
Response: Thanks for identification of this ommission. We have now addressed it:” Setting: Over 3,600 teaching and non-teaching hospitals in 22 countries. We searched CINAHL, The Cochrane Library, PubMed, sciELO, Scopus and Web of Science from the beginning of records until May 2020. Participants: Women at birth. Over 18.5 million births.” (Line 25-27)

Please note that we don’t put a precise estimate of number of hospitals but only the likely minimum as we have more than one study from countries like United States and France and if we add up totals, we may end up double counting hospitals. As same hospitals may be included in different studies of specific country and if we do a simple add up from different studies, we will very much likely double count the hospitals.

Comment: In the conclusion session, the phrase "Nevertheless, teaching hospitals, in several subgroups, have a lower or higher likelihood of cesarean sections" are more or less obvious after the main conclusion and seems useless.

Response: Thank you for this comment. We think that is important to highlight that as in many instances (i.e. country level, risk groups) the odds for CS are lower among teaching hospitals. This is major and important finding in subgroup analysis and the implications of such findings are important to be considered with future research.

Comment: Introduction, pages 4-5: well structured, however, I would recommend introducing the topic of the definition of teaching hospital also here. In addition, another argument regarding the possible characteristics of teaching hospitals should be better explored, on the more appropriate adherence to evidence-based guidelines or clinical protocols in these settings.
Response: Thank you for these suggestions. As suggested by editor, we prefer though to keep the details of teaching definition and groups we have created for analysis in methods section. (Line 139-148).

We have also expanded on adherence to evidence-based guidelines: “Teaching hospitals are generally associated with a higher standard of care and better outcomes.26-32 There is more adherence to evidence-based guidelines and/or clinical protocols. Higher quality care, in turn, can also translate to a lower likelihood of caesarean delivery as a result of fewer unnecessary procedures.” (Line 88-91)

Comment: Material & Methods, page 5-7: As already said, the definition of teaching hospital should come first. There is no information at all on language restriction. How was this treated? I realized that some other databases were not checked like for instance ScieLO.
Response: We have addressed now teaching hospital definition and created groups based on that. We have reviewed again all papers to make sure we have classified all of them according to this new classification.

With regards to language, sorry for our omission. In search strategy we have not put any limitation of language. During review of papers (screening and full text review) we have only reviewed papers that have provided information in English. We clarified this now in the manuscript: “Studies were excluded in case they were not reported in English language.” (Line 113)

We have checked for most important data bases that we think would capture most of literature out there. We think databases that we have chosen would entail most of papers published.

Out of curiosity, and in attempt to address your comments, we have also reviewed sciELO database. With simple search using search terms “caesarean section” and “hospital” we have found 651 papers. We have screened titles and abstract. We have not found additional papers that match our criteria.

Comment: On page 5, line 52, it is said "Studies were excluded to remove overlapping data" and this is not very clear. Does it refer to the situation when more than one published article reported the same dataset of cases? Please clarify.
Response: Apologies for lack of clarity here. Correct. This referrers to the situation where more than one identified article (during screening/full text review) seems to have used same data sets. We check this carefully by looking at period of data collection, population under study and data sets studies have used to confirm this. Then from overlapping studies we would chose studies based on following criteria: studies that report outcomes that are focus of meta-analysis, studies that report adjusted estimates, number of adjusting variables, size of sample size study used, inclusion and exclusion criteria. To address this omission in clarity we have included following: “This refers to the situation where more than one identified article (during screening/full text review) seems to have used same data sets.” (Line 111-113)

Comment: Discussion and Conclusion: they are in accordance with what was planned and presented. Please consider the comments above to adapt them accordingly.

Response: Many thanks for your comment. We have tried to address the comments above which have also affected the discussion section. The changes are visible in manuscript with track changes.

Reviewer: 2
Reviewer Name: Mohammed Walid Zimmo
Institution and Country: Department of Obstetrics and Gynecology, Al Shifa medical complex, Gaza, Palestine
Competing interests: None declared

Comment: Thank you for the opportunity to review this manuscript. It is very interesting to compared the odds of a caesarean section in teaching hospitals compared with non-teaching hospitals. I commend the authors for this important contribution.
Response: Thank you for your kind comments.

Commnet: This is an interesting and important topic, and a well-performed study covering a large population in many countries over an extended period of time could add to the understanding of changes in obstetric practice, and possible strategies to avoid unnecessary cesarean sections. The authors had chosen a significant methodology and well presenting the results.
Response: Thank you for your kind comments.

Comment: I have small comment in the introduction lines 10, authors state
“The WHO in 1985 suggested an optimal rate of 10-15%"
This recommendation was done by WHO in 1983 and they change it in 2015. WHO recommends delivery by cesarean section when clinically indicated.
Please revise this reference:
https://apps.who.int/iris/bitstream/handle/10665/161442/WHO_RHR_15.02_eng.pdf;jsessionid=DB7E8077D62D77B43FEB6BA19049A60F?sequence=1
Response: We have addressed this comment: “Despite recommendations by the WHO to decrease CS rates and perform CS based primarily on medical indications,3-5 CS rates continue to increase, rising to an estimated 21.1% worldwide in 2015.6” (Line 70-72)

Comment: I agree with your discussion specially with your statement "In order to improve the quality of healthcare provision for deliveries, the interplay of factors associated with teaching status of hospitals must be examined, as well as outcomes, related to caesarean section rates but also other outcomes such as proportion of morbidity, mortality, poor-perinatal outcomes and particular forms of delivery (i.e. emergency and primary caesarean, vaginal birth after caesarean)" since we miss sufficient studies which explore proportion of morbidity, mortality, poor-perinatal outcomes.
Response: Thank you for these comments.

VERSION 2 – REVIEW

REVIEWER	Jose Guilherme Cecatti Department of Obstetrics and Gynecology School of Medical Sciences The University of Campinas, Brazil
REVIEW RETURNED	28-Dec-2020

GENERAL COMMENTS	The authors properly addressed the comments arising from the first version of the manuscript and performed the required changes accordingly.